such as neutral, upwards and downwards so, the following movements can be obtained.

Hand feed to the carriage

Keep the feed check lever (F) in neutral position and if cross feed wheel is rotated by hand, the cross slide moves towards the lathe axis.

Automatic feed to the cross slide

When feed check lever is kept in downward (D), gear G2 will be connected to G6. The rotation of G6 will make the cross slide to move automatically.

Automatic feed to the carriage for thread cutting

When the two half nuts in the apron are connected to the leadscrew by using lever (L), leadscrew makes the carriage to move automatically and cut threads for required pitch value.

Lead screw

The lead screw is a long threaded shaft. It is used for moving the carriage automatically to a calculated distance only when threads have to be cut. Mostly leadscrews have an Acme thread.

The leadscrew is held between two bearing infront of the lathe bed. A gear is attached to the leadscrew at left end. The gear is called as gear on leadscrews. A half nuts lever is provided in the apron to engage the halfnuts with the leadscrew.

If the half nuts are engaged with the running screw the carriage will move automatically. When the lead screw is kept stationary the carriage is locked at the required position by using half nuts.

Feed Rod

Feed rod is placed parallel to the leadscrew infront side of the bed. It is a plain long shaft which has a keyway along its length. The power is transmitted from the spindle to the feed rod through tumbler gears and a gear train. It is useful for providing feed movement to the carriage and cross slide, except thread cutting time.

1.8 Spindle Mechanism

The spindle is located in the headstock and it receives the driving power from the motor. The spindle speed should be changed to suit different machining conditions like type of material to be cut, the diameter and the length of the work, type of operation, the type of cutting tool material used, the type of finish desired and the capacity of the machine. In order to change the spindle speeds, any one of the following methods are employed.

1. Stepped cone pulley mechanism
2. Back gear mechanism
3. Gear box mechanism

1.9 Stepped Cone Pulley Mechanism

It is simple in construction. A stepped cone pulley is attached with the spindle contained within the headstock casting. The cone pulley has four steps (A, B, C & D). Another cone pulley having four steps (E, F, G and H) is placed parallel to the spindle cone pulley. Both the cone pulleys are connected by a flat belt. The belt can be arranged between the steps A & H, B & G, C & F and D & E. The cone pulley at the bottom is connected to the electric motor by a 'V' – belt. So the cone pulley at the bottom rotates at a particular speed.

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